

Orbit Wars — Critic-Free RL Math, v12

Four variance-reduction tricks for the GRPO trainer,

and why the un-patched group baseline collapsed

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This note specifies the four changes added to the v12 *critic-free* (GRPO) trainer, the math behind each, and the failure they fix. It is a companion to `rl_math_v9.tex` (the policy, reward, league, and PPO + GAE update are defined there and not repeated). Each trick is gated by a **Config** flag and **defaults to the legacy behaviour**, so a run is bit-for-bit unchanged until a flag is flipped; ablate them one at a time. The four:

tag	flag	idea
[A]	GRPO_ANALYTIC_ADV	closed-form 2p advantage at a shrunk win-rate (Section 2)
[B]	GRPO_STD_NORM, GRPO_LOO	drop the per-group std; leave-one-out baseline (Section 3)
[C]	GRPO_PHI_VALUE	use the shaping potential as a value, not a reward (Section 4)
[D]	GRPO_CRN	common random numbers across a world-group (Section 5)

Contents

1	The critic-free estimator and the collapse it produced	1
2	[A] The 2p advantage in closed form, and its shrinkage	2
3	[B] Drop the per-group standard deviation (and use leave-one-out)	3
4	[C] The shaping potential is a value, not a reward	4
5	[D] Common random numbers across a world-group	5
6	Flags, defaults, and interactions	5

1 The critic-free estimator and the collapse it produced

Variable reference — this section	
B	parallel envs per iteration, $B = n_W \cdot G$
G	<i>group size</i> : rollouts sharing one world (GRPO_GROUP); $n_W = \text{groups/iter}$
w_g	the world (initial planets/seating) shared by all G members of group g
R_i	rollout i 's scalar return (full shaped return, or terminal-only if GRPO_OUTCOME_ONLY)
O_i	terminal outcome of rollout i ; in 2p $O_i \in \{-1, 0, +1\}$, in 4p the placement-linear value
μ_g, σ_g	within-group sample mean / std of $\{R_i\}$
$\hat{A}_i$	the advantage assigned to <i>every</i> timestep of rollout i
ε	GRPO_ADV_EPS, the std floor (10^{-4})
π_θ	the policy; a_t^i, s_t^i the action/state at step t of rollout i

GRPO replaces PPO's learned value baseline with a **group baseline**. Each iteration draws n_W worlds and rolls each out G times (the rollout lays the B envs out as contiguous blocks of G , one world per block, so a block is a group). With the return R_i standardized inside its group,

$$\hat{A}_i = \frac{R_i - \mu_g}{\sigma_g + \varepsilon}, \quad \mu_g = \frac{1}{G} \sum_{j \in g} R_j, \quad \sigma_g = \sqrt{\frac{1}{G} \sum_{j \in g} (R_j - \mu_g)^2}, \quad (1)$$

the gradient is the clipped surrogate of the PPO section applied with $\hat{A}_i$ *flat over all timesteps* of rollout i : $\hat{g} = \mathbb{E}[\sum_t \nabla_\theta \log \pi_\theta(a_t^i | s_t^i) \hat{A}_i]$. There is no critic, no GAE, and (with GRPO_KL_COEF= 0) no reference anchor.

The collapse. A warm-started run climbed to $\text{elo2} \approx 2110$, then over ~ 40 iterations decayed to win-rate 0.09 with the launch rate sliding $0.80 \rightarrow 0.30$ — the project's familiar *passivity basin*. The trigger was an asymmetric outcome decay (a small “lose slower” term), but the *amplifier* was Eq. (1) itself: in a group that loses every rollout, σ_g collapses and the ε -floored division blows a trivial within-group tiebreak up to a full-magnitude “stall” gradient. Sections 2–3 dissect and remove that amplifier; Section 4 restores the per-step credit assignment GRPO structurally lacks; Section 5 tightens the baseline. All four are properties of *this* MDP: a terminal-dominated, near-binary reward evaluated on shared-world groups under self-play.

2 [A] The 2p advantage in closed form, and its shrinkage

Variable reference — this section	
B_o	outcome magnitude in return units, $B_o = \text{WIN_BONUS}/\text{PPO_REWARD_SCALE}$ ($= 10$)
w	number of <i>wins</i> among the G rollouts of a group
$\hat{p}$	empirical group win-rate w/G
$\tilde{p}$	Beta-shrunk win-rate, Eq. (4)
α	GRPO_ANALYTIC_ALPHA, the Beta pseudocount
$\hat{A}_{\text{win}}, \hat{A}_{\text{loss}}$	the two advantage values a 2p group can take

In 2p the outcome is binary, so (ignoring draws and any decay) every $R_i \in \{+B_o, -B_o\}$. A group with w wins, $\hat{p} = w/G$, has *exact* statistics

$$\mu_g = B_o(2\hat{p} - 1), \quad \sigma_g = 2B_o\sqrt{\hat{p}(1 - \hat{p})}, \quad (2)$$

and substituting into Eq. (1) (with $\varepsilon \rightarrow 0$) the standardized advantage takes **only two values**, a pure function of the win-rate:

$$\boxed{\hat{A}_{\text{win}} = \sqrt{\frac{1-\hat{p}}{\hat{p}}}, \quad \hat{A}_{\text{loss}} = -\sqrt{\frac{\hat{p}}{1-\hat{p}}}} \quad (3)$$

Derivation. $\hat{A}_{\text{win}} = \frac{B_o - \mu_g}{\sigma_g} = \frac{2B_o(1-\hat{p})}{2B_o\sqrt{\hat{p}(1-\hat{p})}} = \sqrt{\frac{1-\hat{p}}{\hat{p}}}$, and symmetrically for the loss. Two consequences:

- **Rare events are up-weighted.** As $\hat{p} \rightarrow 0$ (a world you almost always lose) a rare win gets $\hat{A}_{\text{win}} \rightarrow +\infty$ while the common losses get $\hat{A}_{\text{loss}} \rightarrow 0^-$: GRPO implicitly chases the rare win. This is a useful exploration prior — when it is real.
- **Every group carries unit gradient energy.** $\hat{p}\hat{A}_{\text{win}}^2 + (1-\hat{p})\hat{A}_{\text{loss}}^2 = (1-\hat{p}) + \hat{p} = 1$ for *all* $\hat{p}$. A 99% blowout pushes the gradient exactly as hard as a 50/50 toss-up.

The collapse lives at $\hat{p} \in \{0, 1\}$: then $\sigma_g = 0$, the division falls back on ε , and whatever infinitesimal residual distinguishes the (all-losing) rollouts — a length-dependent decay term, sampling noise — is renormalised to magnitude 1. The estimator manufactures a confident gradient out of noise.

The fix: evaluate Eq. (3) at a shrunk $\tilde{p}$. Replace the empirical $\hat{p}$ by a $\text{Beta}(\alpha, \alpha)$ posterior mean,

$$\tilde{p} = \frac{w + \alpha}{G + 2\alpha} \in (0, 1), \quad (4)$$

and *assign Eq. (3) by outcome class* (wins $\rightarrow \hat{A}_{\text{win}}$, losses $\rightarrow \hat{A}_{\text{loss}}$, draws $\rightarrow 0$). This (i) keeps $\tilde{p}$ strictly inside $(0, 1)$, so there is no ε -corner; (ii) gives *every win in a group the same advantage and every loss the same advantage*, so any within-class tiebreak (the stall signal) is gone by construction; and (iii) is a **Rao–Blackwellisation** — the outcome class is the sufficient statistic for a binary reward, and conditioning on it replaces the noisy per-sample value with its exact conditional advantage, which is variance-optimal among unbiased estimators that respect the outcome. In 4p the outcome is graded (placement-linear), Eq. (3) no longer holds, and the code falls back to [B].

3 [B] Drop the per-group standard deviation (and use leave-one-out)

Variable reference — this section

b_g	a per-group baseline subtracted from R_i
$\mu_{g,-i}$	leave-one-out group mean, Eq. (6)
$\mathcal{E}_g$	per-group gradient <i>energy</i> $\frac{1}{G} \sum_{i \in g} (\text{advantage})^2$
$p(1-p)$	the PFSP-“even” matchmaking weight (<code>rl_math.v9.tex</code> , league section)

For a constant-per-rollout advantage the variance-minimising baseline is the expected return, $b_g = \mathbb{E}[R \mid w_g] \approx \mu_g$; subtracting it is correct. **Dividing by σ_g is not part of any unbiased estimator** — it is a per-group reweighting, and Eq. (3) showed exactly what it does: it forces every group to unit energy. Compare the per-group energy with and without it:

$$\underbrace{\mathcal{E}_g / \sigma^2 = 1}_{\text{current (DeepSeek)}} \quad \text{for all } \hat{p}; \quad \underbrace{\mathcal{E}_g^{\text{center}} = \sigma_g^2 = 4B_o^2 \hat{p}(1-\hat{p})}_{\text{center-only (Dr. GRPO)}}. \quad (5)$$

The center-only energy is **peaked at $\hat{p} = \frac{1}{2}$ and vanishes at the extremes** — it down-weights blowout/hopeless groups (nothing to learn) and concentrates the gradient on contested ones. This is precisely the right curriculum, and it is *the same function* $p(1-p)$ that PFSP-“even” uses to choose opponents. The league is already sampling matchups to maximise $p(1-p)$; the $/\sigma_g$ term flattens that back out, so matchmaker and update were optimising against each other. Removing it aligns them.

Implementation. `GRPO_STD_NORM=False` skips the division and applies *one global scale* to the batch (a single std over all transitions), which — unlike the per-group σ_g — preserves the cross-group $p(1-p)$ weighting while keeping the overall step size $O(1)$.

Leave-one-out baseline (RLOO). `GRPO_LOO=True` replaces μ_g by the mean of the *other* $G-1$ rollouts,

$$\mu_{g,-i} = \frac{1}{G-1} \sum_{j \in g, j \neq i} R_j \implies R_i - \mu_{g,-i} = \frac{G}{G-1} (R_i - \mu_g). \quad (6)$$

The baseline no longer contains sample i , removing the $O(1/G)$ self-correlation that biases the included-mean estimator; the identity on the right shows the only effect on the *magnitude* is a benign $G/(G-1)$ rescale (here 16/15), so the value of LOO is the de-biasing, not a scale change.

4 [C] The shaping potential is a value, not a reward

Variable reference — this section	
$\Phi(s)$	shaping potential, $\Phi = \text{SHAPE_SHIP } \varphi_{\text{ship}} + \text{SHAPE_PROD } \varphi_{\text{prod}}$
$\varphi_{\text{ship}}, \varphi_{\text{prod}}$	log ship-margin / production-share vs. the strongest opponent (Ng potentials)
γ, λ	discount (GAMMA) and GAE trace (GAE_LAMBDA)
$V_t, \hat{V}_t$	a value at s_t ; the hand-built estimate Eq. (8)
a	fitted slope of outcome on terminal position change, Eq. (9)
$\delta_t, \hat{A}_t$	TD residual and the per-step (GAE) advantage
L	rollout length; $\Delta\Phi_t = \Phi(s_t) - \Phi(s_0)$

GRPO’s deepest weakness is that one trajectory-level scalar is smeared over all $T=500$ steps: no *per-step* credit. We already compute a hand-crafted “who is winning” signal $\Phi(s)$ and feed it as potential-based reward shaping (PBRS). Under the Monte-Carlo return GRPO actually uses, that is nearly wasted. Telescoping the shaping increment $\gamma\Phi_{t+1} - \Phi_t$ over a rollout,

$$\sum_{t=0}^{L-1} (\gamma\Phi_{t+1} - \Phi_t) = \underbrace{-\Phi_0}_{\text{shared in group}} + (\gamma-1) \sum_{t=1}^{L-1} \Phi_t + \gamma\Phi_L. \quad (7)$$

The leading term $-\Phi_0$ is the world’s *initial* imbalance — identical for all G members of a group (they share w_g , hence s_0) — so the group mean μ_g **cancels it exactly**. What survives is a tiny $(\gamma-1) = -0.003$ -weighted occupancy integral, not the credit signal we wanted. This is the textbook fact (Ng et al.; Wiewiora 2003) that *PBRS is equivalent to initialising the value function to Φ , and only does work in the presence of bootstrapping*. With no critic there is no bootstrapping, so the shaping is inert.

The fix: bootstrap with Φ . Use Φ as the value and run GAE. Anchor the value at the group baseline and let position change move it:

$$\widehat{V}_t^i = \mu_{g(i)} + a(\Phi(s_t^i) - \Phi(s_0^i)) = \mu_g + a\Delta\Phi_t^i, \quad (8)$$

with the slope fit once per iteration by closed-form least squares of the group-centred outcome on the *terminal* position change $\Delta\Phi_L$ (clamped to $[0, \text{GRPO_PHI_A_MAX}]$):

$$a = \frac{\sum_i (R_i - \mu_g) \Delta\Phi_L^i}{\sum_i (\Delta\Phi_L^i)^2}. \quad (9)$$

Then standard GAE on the (now sparse: terminal-outcome-only) reward r_t ,

$$\delta_t = r_t + \gamma\widehat{V}_{t+1} - \widehat{V}_t, \quad \widehat{A}_t = \sum_{l \geq 0} (\gamma\lambda)^l \delta_{t+l}. \quad (10)$$

Two things fall out. At the root $\widehat{V}_0 = \mu_g$, so δ_0 subtracts the group baseline: **world difficulty is cancelled by the shared root**, recovering GRPO’s whole purpose. And for an interior step ($r_t = 0$, $\gamma \approx 1$),

$$\delta_t \approx a(\Phi(s_{t+1}) - \Phi(s_t)), \quad (11)$$

so the per-step advantage credits exactly the moves that *improve your position*, with the terminal outcome propagated back through them by the $(\gamma\lambda)$ trace. Because $\widehat{V}$ is a function of state only, it is an *unbiased baseline* regardless of fit quality (a bad fit costs variance, never bias); the $\lambda < 1$ trace is the usual bias–variance knob. To avoid double-counting Φ , the explicit reward shaping is turned off in this mode (GRPO_PHI_VALUE routes Φ to the value, and the dense reward is set to zero).

5 [D] Common random numbers across a world-group

Variable reference — this section	
ξ_i, ζ_i	the <i>ego</i> ’s and the <i>opponent</i> ’s action-sampling noise in rollout i
w_g	the (shared) world; comets are deterministic per world
u, g_u	a uniform / Gumbel sample driving the categorical and the gate
Cov	covariance; the CRN variance identity below

A rollout’s return is a function of three independent noise sources, $R_i = f(w_g, \xi_i, \zeta_i)$: the world (shared within a group), the ego’s sampling ξ_i , and the neural opponent’s sampling ζ_i . The group baseline is meant to isolate the effect of varying the *ego*’s actions, but with independent ζ_i it also absorbs opponent-sampling noise, inflating σ_g and loosening the advantage. **Common random numbers** couple the opponent across the group: draw one noise stream ζ_g and share it among all G members (the ego keeps independent ξ_i — that variation is the signal). By the control-variate identity

$$\text{Var}(R_i - R_j) = \text{Var } R_i + \text{Var } R_j - 2 \text{Cov}(R_i, R_j), \quad (12)$$

raising Cov between groupmates (via shared ζ) lowers the variance of within-group differences, hence of $\widehat{A}_i$. Concretely the categorical **WHERE** is sampled by Gumbel-max with noise shared across the group, and the **GATE** by a shared uniform:

$$d_s^i = \arg \max_d (\ell_{s,d}^i + g_u[g(i), s, d]), \quad f_s^i = \mathbb{I}[u[g(i), s] < p_s^i], \quad (13)$$

where g_u, u are indexed by the *group* $g(i)$, not the rollout — identical within a group, independent across groups. Identical-logit groupmates therefore sample identical actions (a clean unit test); when logits differ (the ego has diverged the state) the actions differ but the *noise* is common, which is the variance reduction. CRN applies only to neural opponents (scripted seats are handled in the engine); the world and its comets are already shared by construction.

6 Flags, defaults, and interactions

flag	default	effect when changed
GRPO_STD_NORM	True	False → center-only + global scale (Dr. GRPO)
GRPO_LOO	False	True → leave-one-out group mean (RLOO)
GRPO_ANALYTIC_ADV	False	True → closed-form 2p advantage at shrunk $\tilde{p}$ (4p falls back)
GRPO_ANALYTIC_ALPHA	1.0	Beta pseudocount in Eq. (4)
GRPO_PHI_VALUE	False	True → Φ -as-value GAE (per-step credit); shaping reward off
GRPO_PHI_A_MAX	3.0	clamp on the fitted slope a , Eq. (9)
GRPO_CRN	False	True → couple neural-opponent sampling within a group

Interactions. [A] and [B] both shape the *flat* trajectory advantage and are mutually exclusive for the terminal term: with [A] on (2p) the analytic values are used and the std/LOO knobs of [B] only toggle the optional global whiten. [C] *replaces* the flat path with the GAE path, so [A]’s analytic values do not apply while [C] is on (the group baseline enters [C] through μ_g in Eq. (8), and GRPO_LOO still selects $\mu_{g,-i}$ there). [D] is orthogonal — it changes how the *rollout* is sampled, not how the advantage is computed — and composes with any of the above. Recommended first ablations: **[B] alone** (cheapest, removes the collapse amplifier and aligns with PFSP) and **[C] alone** (the per-step credit, the largest expected upside); then [C]+[D].